GE Medical Systems

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Technical Publication

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Revision 02

GE Medical Systems
HiSpeed QX/i Linux™ Software Installation Procedures

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- **THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.**
- **IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.**
- **DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.**
- **FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.**

AVERTISSEMENT

- **CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.**
- **SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.**
- **NE PAS TENTER D'INTERVENTION SUR LES éQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS éTé CONSULTé ET COMPRIS.**
- **LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAîNER CHEZ LE TECHNICIEN, L'OPéRATEUR OU LE PATIENT DES BLESSURES DUES à DES DANGERS éLECTRIQUES, MéCANIQUES OU AUTRES.**

WARNUNG

- **DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.**
- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENöTIGT, IST ES AUFGABE DES KUNDEN FüR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERäT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLäGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

AVISO

- **ESTE MANUAL DE SERVICIO SóLO EXISTE EN INGLéS.**
- **SI ALGúN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLéS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIóN.**
- **NO SE DEBERá DAR SERVICIO TéCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELéCTRICAS, MECáNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTêNCIA TéCNICA Só SE ENCONTRA DISPONíVEL EM INGLêS.**
- **SE QUALQUER OUTRO SERVIçO DE ASSISTêNCIA TéCNICA, QUE NãO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, é DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIçOS DE TRADUçãO.**
- **NãO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTêNCIA TéCNICA.**
- **O NãO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANçA DO TéCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELéTRICOS, MECâNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE è DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE è TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

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注意:

本维修手册仅存有英文本。
非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
未详细阅读和完全了解本手册之前，不得进行维修。
忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write “Damage In Shipment” on ALL copies of the freight or express bill BEFORE delivery is accepted or “signed for” by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE’s electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE’s field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION Risk of Explosion **Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.**

ATTENTION Danger d'Explosion **Il y a danger d'explosion s'il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.**

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
0	05/08/03	Initial Release of Document.
1	03/24/04	Updated Section 2.1 for HiSpeed QX/i M4 software and Section 2.2 and 3.5 for CSA protocols save/restore. Combined sections 3.1 and 3.2 (Install Operating System) and eliminated duplicate step, updated Figure 1-4 (System Settings).
2	06/02/04	Chapter 1, Section 2.1.2 M4 Software CDs: Corrected the CD P/N.

Table of Contents

Preface
Publication Conventions

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0
General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles and shading are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the chapter, section or subsection, Such as used at the beginning this chapter for example.

Note: **Conveys information that should be considered important to the reader.**

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents "additional" information that may or may not be relevant.

Section 2.0
Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys — A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Soft Keys — Whereas the computer MENU button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.

Section 3.0
Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: Fixed Output

`This paragraph denotes computer screen fixed output. It’s output is fixed from the sense that it does not vary from application to application.It’s the most commonly used style used to indicate filenames, paths and text.`

Example: Variable Output

This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>

Example: Fixed Input

This paragraph denotes fixed input. It’s typed input that will not vary from application to application. Fixed text the user is required to supply as input.

Example: Variable Input

This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.

Chapter 1
Software Load Procedure

Section 1.0
Introduction

1.1 Document Overview

1.) Before you begin, become familiar with the representation of computer text and character used in the load from cold procedure. Please review the Preface - "Publication Conventions", section Section 3.0 on page 14. Chapter 1 is organized as follows:
   - Section 1.0 - "Introduction": Overview of the SW installation procedure.l
   - Section 2.0 - "Before You Begin": Things you should know and do before you begin.
   - Section 3.0 - "Load Procedure" for the operating system and CT applications software.

2.) It may be necessary to reference the following information in this document during the installation procedure, pending your specific needs:
   - Chapter 2 - "Re-configuration of CT Application SW"
   - Chapter 4 - "Changing System Time Zone"
   - Chapter 5 - "Regenerating a Scan Database"
   - Chapter 3 - "Laser & DICOM Camera Setup"
   - Appendix A - "Error Messages And Troubleshooting"

1.2 Software Load Procedure (LFC) Overview

The CT software load procedure (a.k.a., “Load From Cold” or LFC) prepares the system disk, installs and configures the operating system, and installs and configures applications software on the OC computer.

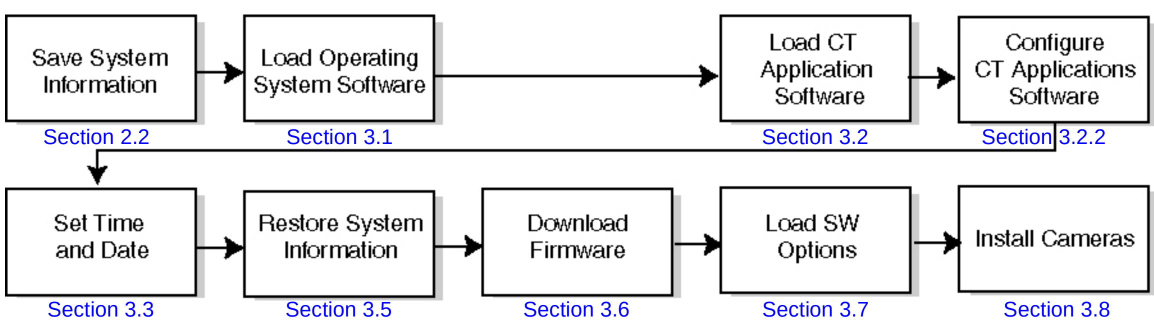

Figure 1-1 Software Overview

Section 2.0
Before You Begin

2.1 Gather Software CD-ROMs

Operating and applications software is contained on two (2) CD-ROMs.

2.1.1 M3 Software CDs

To load M3 software, you must have the following CD-ROMs:

1.) Linux Application: 2395156
2.) Linux™ OS ISO Image: 2392921

2.1.2 M4 Software CDs

To load M4 software , you must have the following CD-ROMs:

1.) Linux Application: **5115372 - HiSpeed QX/i M4 Applications Software**
2.) Linux OS: **5115543 - LightSpeed/HiSpeed Linux Console Operating System**

Note: For M4 software, DO NOT USE the **5115546 - LightSpeed/HiSpeed QXI DARC Apps M4** CD or **5115552 - Xtream Serial-Over-Lan Service Software** CD. They are only used on GRE Consoles.

2.2 Save System State to DVD and CSA Protocol Retention

2.2.1 Save System State

Before beginning the software install, you should save the system configuration information, characterization, calibration, protocols, etc. to a System State DVD.

1.) Bring the system up if it is not already up.
2.) Insert the System State DVD.
3.) Click on the SERVICE DESKTOP.
4.) Click on the PM icon.
5.) Click on SYSTEM STATE.
6.) Click on ALL. This will highlight Cals, Characterization, Reconfig Info, etc.
7.) Click on SAVE.
The save will take a few minutes. Review the output for errors or missing files; the scroll bar on the right works only when the tool isn't busy performing some task, it may take a little while. If you see any missing files or failures, then refer to the note below.
8.) (Optional) To restore Protocols from a previous system, Applications must be up. Bring up Applications, then perform the following two steps:
a.) Open a shell and type:
`sysstategui -MOD -IRX2LNX` ENTER
b.) When the pop-up window appears, select PROTOCOLS, then RESTORE.

9.) Click on DISMISS.

If System State reported no problems, then you may proceed to Section 2.2.2

Note: **IF YOU SHOULD GET SAVE/RESTORE SYSTEM STATE ERRORS:** The Save/Restore System State process reports status on each file specification listed in the `/usr/g/config/SysState.cfg` file. The status may be:

- `Save (or Restore) of {filename} succeeded`
- `Save (or Restore) of {filename} skipped. File not found`
- `FAILED`

A status of `FAILED`, indicates potential problems, `skipped` means the file not found was not required. Please review the following for each file you see an error reported.

Some files simply may not exist on the system. If certain tools were never run then certain files may never get created. You can view `/usr/g/config/SysState.cfg` and look for a required field. `req=Y` means the file in question is required. You can also check the system and DVD to verify whether it was actually missing or if there were permission or owner problems with the file.

2.2.2 CSA Protocol Retention

Sites performing a Software Upgrade or a Software Reload should perform this sub-section to manually retain their CSA Protocols (Reformat, Volume Viewer, etc.).

Software Reload: Follow the instructions below for the following software reload:

- GOC2/Linux Console M3 software to GOC2/Linux Console M3 software

Software Upgrade: Follow the instructions below for the following software upgrade:

- GOC2/Linux Console M3 software to GOC2/Linux Console M4 software

1.) Open up a Unix Shell.

2.) Verify the System State DVD is in the Peripheral Tower DVD RAM drive.

3.) Type the following to save CSA Protocols:

a.) **`tar`** SPACE **`Pcvf`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/usr/g/ctuser/Prefs/VoxtoolPrefs/*`** ENTER

b.) **`mountDVD`** ENTER

`Mounting DVD.`

c.) **`\cp`** SPACE **`-f`** SPACE **`/data/vxtlBackup.tar`** SPACE **`/DVD/service_mod_data/system_state`** ENTER

d.) **`unmountDVD`** ENTER

`Unmounting DVD.`

2.3 Record KEY System Information

Keep a record of system information in a safe place. System information is saved on the System State DVD, but to be safe it should also be recorded somewhere else at your site in case the State DVD becomes lost or damaged.

1.) Record the most important Reconfig INFO for the scanner. Use the following procedure to help gather and record key system information from the INFO file and other system files.

a.) Open a shell on the OC.

b.) View the following files to find the information for the chart on the following page (Table 1-1); type the following:

`more /usr/g/config/INFO` ENTER

Gets all data except Exam Number and Camera information. HIS/RIS information will not be in the INFO file if the customer does not have this option.

`more /usr/g/bin/scanRx.db` ENTER

Gets the next Exam Number (Add 1 to the number to record the "next" exam number.)

For each of the fields below look at the `more file output` to find the associated **`setenv`** value and record the value in Table 1-1.

Note: IP Addresses and Net/Broadcast Masks have the following format: ***www.xxx.yyy.zzz***

NOTICE Potential for Data Loss **Be certain to perform a Safety Backup.**

SYSTEM NETWORK CONFIGURATION

	FIELD NAME:	"setenv" NAME:	VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number**	Use procedure above to determine Exam Number.	
	DASM Type	DASMTYPE	See Chapter 3
	Camera Type	CAMERATYPE	See Chapter 3
Network Settings:	Host Name	GATEWAY_HOSTNAME	
	Gateway IP Address	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Default Gateway	GATEWAY_DEFAULT	
	Suite Name	SUITEID	
Optional - Network Printer	Printer Hostname	PRINTER_HOSTNAME	
	Printer IP Address	PRINTER_IPADDRESS	
Optional - HIS/RIS Systems*	CT AE Title	HRCTTITLE	
	Port Number	HRPORTNUM	
	H/R Server AE Title	HRAETITLE	
	IP Address	HRIPADDR	

Table 1-1 Re-configuration Information

Note: * HIS/RIS information is not in the INFO file if the customer does not have the HIS/RIS option.

** Do not record the Exam Number from the data contained in the INFO file, because the value will be from the last time reconfig was performed. (See the procedure on the previous page to determine the current Exam Number.)

c.) Go to Chapter 3 - "Laser & DICOM Camera Setup", Table 3-1 on page 53 or Table 3-4 on page 54 to record camera information in the appropriate table provided.

2.) Now record the Networking Application (Image transfer) Configuration information, see Table 1-2. On the ImageWorks Desktop, in the Browser under Network, for each host configured for networking do the following and record the information in the table provided:

a.) Click on IMAGE WORKS

b.) Click on NETWORK

For each host configured for networking, perform the following steps, and record the information in Table 1-2 on page 20

c.) Click on SELECT REMOTE HOST.

d.) Click on the HOSTNAME.

e.) Click on UPDATE.

f.) Write the information in the table provided on the next page.

g.) Click on CANCEL to exit Select Remote Host.

If you find this process tedious, you can look at the file by typing:

`more /usr/g/ctuser/Prefs/SdCRHosts` on the system to get this information. The `1` in the file means Advantage Net, the `2` means DICOM. If you see the pound sign (#), it's used as a separator and means nothing.

TYPICAL OUTPUT WOULD APPEAR AS THE FOLLOWING:

A	B	C	D	E	F	G	H	I	J
3.7.52.135	engbay04	#2	4006	engbay04	No comment	0	1	1	0
3.7.52.151	engbay26	#2	4006	engbay26	No comment	0	1	1	0
3.7.52.121	engbay16	#1	4005	engbay16	No comment	0	1	0	1

The fields in the above file relate to the Network GUI settings.

A - IP Address

B - Hostname

C - # is a space, 1 is Advantage Net Protocol, 2 is DICOM Protocol

D - Port Number

E - AE Title

F - Comment Fields

G - Archive Node - 1 = Yes, 0 = No

H - Send Images - 1 = Yes, 0 = No

I - Query/Retrieve - 1 = Yes, 0 = No

J - Custom Search - 1 = Yes, 0 = No

3.) Check the memory to ensure it is recognized. Open a shell and type: **`hinv`** ENTER.

The output for the OC memory should be at least 2048MB. If it is not, check the Host computer to make sure the proper amount of memory is installed

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION					**	**	**	**	**
Host name	Network address	Network protocol	Port number	AE Title	Archive Node	Send Images	Query/ Retrieve	Custom Search	Comments

Table 1-2 Network Application Configuration

Section 3.0
Load Procedure

NOTICE Potential for Data Loss **Make sure that all Images have been reconstructed and archived before performing this procedure. This procedure will re-initialize all system data disks, erasing all images and scan data.**

3.1 Install Operating System

1.) Click on the SHUTDOWN icon (shown in Figure 1-2). An attention box will pop up asking whether to restart or shutdown the system.

2.) Select SHUTDOWN, then click OK. This will shutdown both system Applications and the OS.

Figure 1-2 Shutdown Icon

The following message should now come up:

`Power Down.`

3.) Remove the Console’s front cover.

4.) Loosen the two screws that hold the computer tray in place.

5.) Pull the tray out to provide access to the HP PC’s DVD-ROM drive.

6.) Insert the Linux Operating System CD-ROM into the HP PC’s DVD-ROM drive.

7.) Power OFF the console.

8.) Power ON the console. The computer will come up, displaying the booting process messages.

9.) After the booting process is completed, it will prompt with:
`boot:`

10.) Type: **`GEMS2`** **ENTER** *(for English Keyboard only)*
or: **`iGEMS2`** **ENTER** *(international command only)*

Note: There is no space between the "GEMS" (or "iGEMS") and the "2"

11.) The Linux OS ISO Image will be loaded. The OS installation procedure will proceed by installing the entire Linux OS.

12.) After the OS is loaded, a red button will appear and display OK. Press **ENTER**, and immediately remove the OS CD, when it is ejected.

13.) The system will reboot.

Note: If the OS ISO Image Disk was NOT removed, eject it now by pressing the eject button on the drive. Then, cycle power to the console.

14.) The system will come up with the X-Windows environment in place. Click OK to proceed and display a shell.

3.2 Install CT Application Software

3.2.1 Starting the Installation

1.) Insert the Linux Applications Software disk into the DVD-ROM drive, on the Host PC.

Note: For M4 software, DO NOT USE the 2394896 - LightSpeed/HiSpeed QXI DARC M4 Apps CD. This CD is only used on GOC3/GRE consoles to load software on the DARC hardware.

2.) A pop-up box displays:

`Do you wish to run /mnt/cdrom/autorun`

Click YES

Note: If the pop-up does not appear, eject the Applications CD and then re-insert it, to try again.

3.) The Install Utility Window (Figure 1-3) will appear.

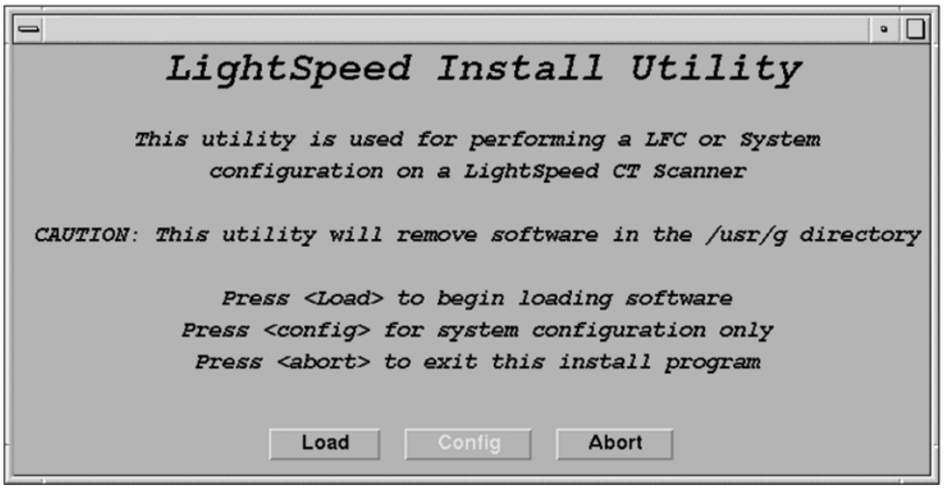

Figure 1-3 Install Utility Window

4.) In the Install Utility window, click on LOAD. A pop up message query is displayed. Asking if you wish to load the INFO file from the System State DVD. If you have a System State DVD, select YES. Another pop-up message appears. Make sure the System State DVD is in the DVD drive in the SCSI Tower (on top of the console) and then select OK.

3.2.2 Configure Application Software

In the steps that follow, go through each of the configuration screens and verify the information is correct for your system. Make chnages as necessary. If you do not have a System State DVD, enter the appropriate information (refer to the data you recorded in section 2.2.2 on page 17).

3.2.2.1 Configure System Settings

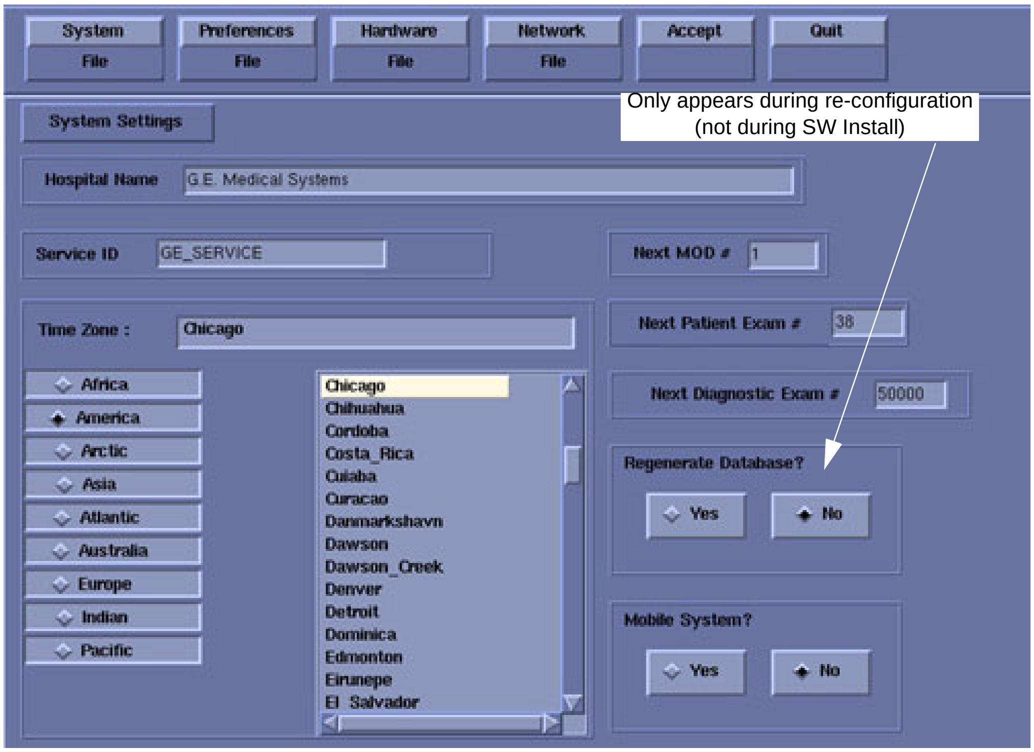

Figure 1-4 System Settings Sample Screen

1.) Change the `Hospital Name` to the site's preferred name.

2.) Enter the `Service ID` issued by the service organization.

3.) Select the proper `Time Zone` values.

Use the scrollbar at the bottom of the timezone selection list to view the entire description of the timezone you are about to select, to ensure that you are selecting the correct timezone for your location.

If the time zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect.

Note: The `Next MOD#` and `Next Diagnostic Exam #` screens are currently not implemented.

4.) Verify and/or change as necessary the `Next Patient Exam #`. The value should be one larger than the "last exam number"recorded. The value entered configures the next Exam number the scan user interface will use. Get this information from the customer's patient log, or by examining information recorded previously.

5.) If you have a mobile CT system, select YES, else the default in NO.

6.) Select PREFERENCES to proceed to the next screen.

3.2.2.2 Configure Preferences

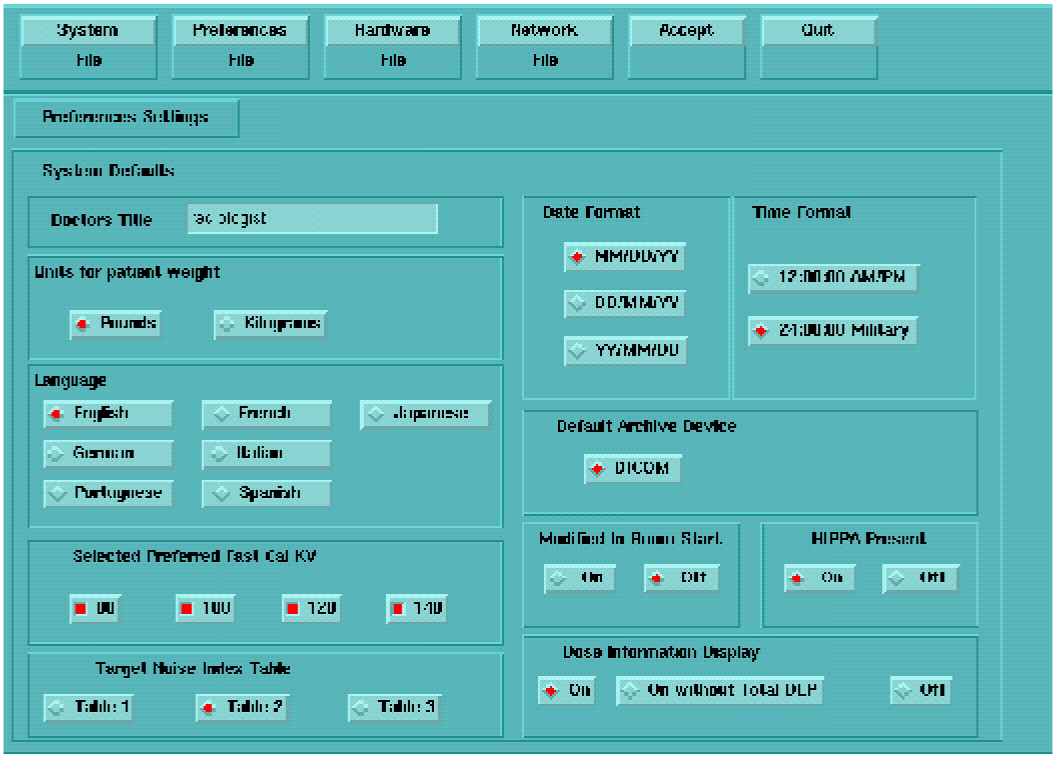

Figure 1-5 Preferences Setup Sample Screen

1.) Select the `Language` (ENGLISH, FRENCH, GERMAN, ITALIAN)to display on the system. See the second Notice in Section 3.7 .

2.) Select `Preferred FastCal KV` - select the kV to be calibrated during FastCal. (Default selections are 120 and 140)

These kVs should include all kVs which the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

3.) Select the desired `Date Format` and the desired `Time Format`.

4.) `Default Archive Device` is DICOM.

5.) The default value for `HIPAA Present` is OFF. When set to ON, this setting enables a login screen on start-up, in which the user needs to enter a user name and password before accessing the system. There is a service user and admin user shipped with the system. The other accounts need to be added, or restored from system state if previously saved. This feature has been added to comply with new federal security regulations. Unless the customer specifically chooses to use this feature, leave it disabled (OFF).

6.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:

- Select ON (full CTDiw Display)
- Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
- Select OFF (no CTDIw Display)

7.) Select HARDWARE to proceed to the screen in the next step.

3.2.2.3 Configure Hardware Settings

Note: Verify hardware — Verify your scan recon hardware and gantry type. If you choose incorrectly now, you will have to re-load software completely.

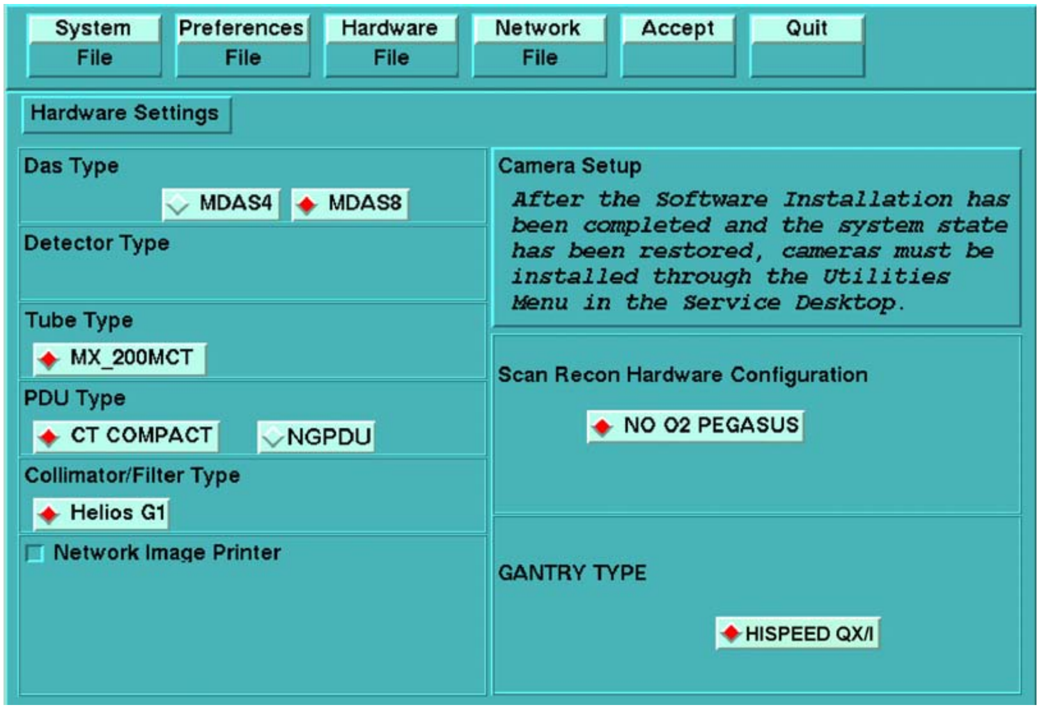

Figure 1-6 Hardware Settings Sample Screen

1.) Select the proper DAS Type for the system:
  - MDAS_H16_16 for LightSpeed16
  - MDAS8 for LightSpeed Ultra
  - MDAS4 for HiSpeed QX/i
  - MDAS4 for LightSpeed Plus w/MDAS
  - SDAS4 for LightSpeed Plus w/SDAS

Note: Figure 1-6 is a sample screen only. The choices displayed for Das Type will vary, according to the system on which the software is loaded. For example, when the software is loaded on a LightSpeed 16, only the MDAS_H16_16 choice will appear.

2.) Select MX_200MCT for the `Tube Type` for the system.

3.) Select NGPDU for the `PDU Type` for the system.

4.) Select HELIOS G1 for the `Collimator/Filter Type`.

5.) Select HISPEED QX/I for `GANTRY TYPE`.

6.) Select NO O2 PEGASUS for the SCAN RECON HARDWARE CONFIGURATION.

7.) Do not enable/select `Network Image Printer` option. Network Image printers must be configured following the installation of system software using the **`reconfig`** utility. Please see the following note.

Note: Print Servers	If system software has already been installed, and your network image printer meets the following criteria, you may enable the option. - The device connected cannot be a print server. For example, a computer with a printer attached. Printers must be connected directly to the network. - The connected device is a true postscript printer (such as Codonics printers) only. When the option is selected, you are asked to supply the following information: - Enter the printer Hostname in the `Name` field. - Enter the printer's IP Address in the `IP Address` field.

8.) Camera setup:

Cameras setup is preformed using utilities accessed through the Service Desktop. See *Section 3.8 .*

9.) Select NETWORK to proceed to the screen in the next step.

3.2.2.4 Configure the Network Settings

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask, and Default Gateway should be obtained from the hospital's network administrator prior to beginning.

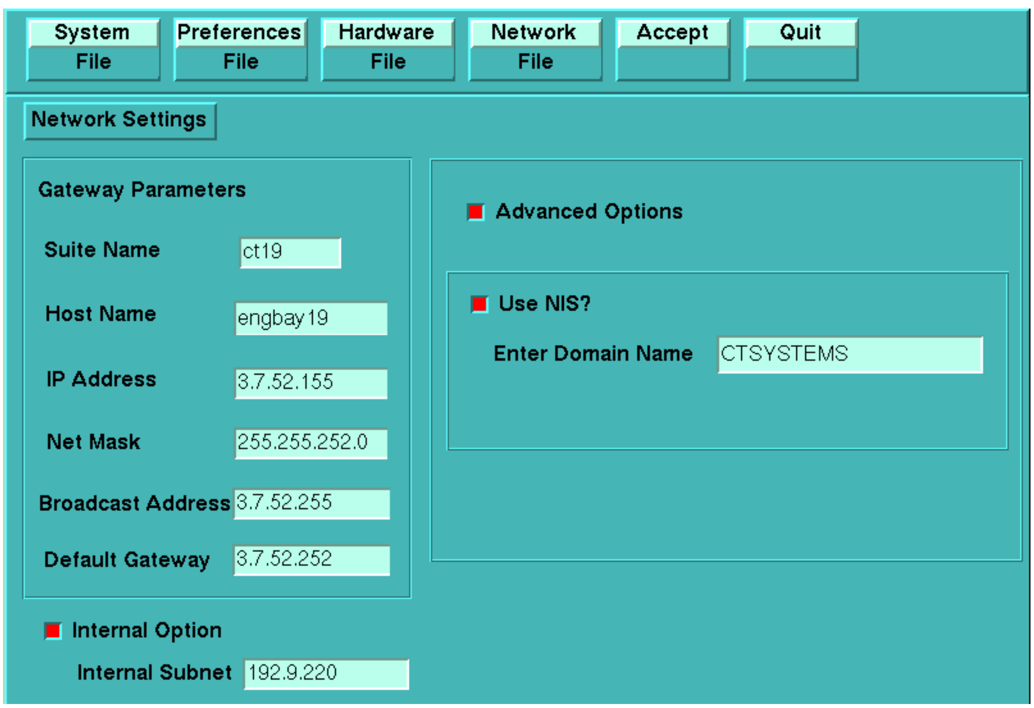

Figure 1-7 Network Settings Sample Screen

Refer to the data you recorded in Section 2.2.2 to verify these entries.

1.) Enter the `Suite Name`.

The `Suite Name` must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is *`<Suite Name>`*`_oc` within the scanner's subnet. It's suggested that you choose CT01 as its name, unless a different Suite Name is required.

2.) Enter the `Host Name`. The `Host Name` identifies the hostname and AE Title of the scanner. The `Host Name`:

- **MUST NOT** be `<Suite Name>_oc` or `<Suite Name>_OC`.
- **MUST NOT** exceed 16 Characters.
- **MUST** only contain the following characters: **`A` through `Z`, `a` through `z`, `0` through `9`, `-` and `_`**

3.) Enter the Host System's `IP Address`.

If possible consult the site's network administrator before configuring the interface to the site's network. The site administrator should provide the System's `Host Name`, `IP Address`, `Net Mask`, `Broadcast Address`, and `Default Gateway` (if a Default Gateway is used).

4.) If you changed the Host System's IP address from the default then the `Broadcast Address` also needs to be updated. The Broadcast Address should be the Gateway IP address with the bits of the host id portion of the address either set to 1s or 0s depending on the configuration of the network. The standard default is 1's but older SunOS machines used 0's.

If the Host System's IP address is 192.100.9.17 the broadcast address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address. If the network contains genesis based scanners or other SunOS 3.5 or 4.1 machines the broadcast address should be 192.100.9.0.

5.) Enter the `Default Gateway` IP Address (optional).

If there is a Default Gateway for the site's network, enter the IP Address of the gateway. Otherwise, leave the field blank.

6.) Select ADVANCED OPTIONS. The Advanced networking options provide support for `NIS`.

Do Not turn on NIS unless requested by your site's network administrator. Turning on NIS by selecting the `Use NIS?` option allows the scanner to use the site's NIS (a.k.a. Yellow Pages) database. The `Domain Name` must be provided by the site's network administrator

---

- To turn on NIS:
  - A.) Highlight ADVANCED OPTIONS.
  - B.) Highlight USE NIS?.
  - C.) Enter ***`Domain Name`***
  - D.) Enter server's ***`IP Address`***
- To Turn off NIS:
  - A.) Highlight ADVANCED OPTIONS.
  - B.) De-select USE NIS.

System is now ready to initiate installation.

3.2.3 Finishing Up

1.) Select ACCEPT to utilize the configuration and continue installation.

The system will automatically reboot.

2.) Click OK, when the pop-up appears. The system will reboot and display a message box.

3.) Click OK, when requested.

Note: Be patient.

This process may be lengthy. The user is advised to exercise patience.

4.) The system will load the CT software, OS patches, kernel changes and configure the system on the OC.

This loading process will take 15-20 minutes.While the load is going on, the results will be displayed in a shell window (which closes when the load is complete). All the window output is logged to a file named `/var/adm/install.log.YYYYMMDDWWWHHMMSS`. (Where *YYYYMMDDWWWHHMMSS* is the Date/Time that the process was started.)

Note: Pflash Failure	During an applications load, the system automatically performs a firmware flash and reconfigures the Scan Data Disk. If either should fail, the system logs (but does not display) the event. Prior to the reboot prompt in the next step, a pop-up window will be displayed, indicating that pflash or re configuration has failed and must be manually performed. Press OK to continue. To perform the manual pflash, see Appendix A - "Error Messages And Troubleshooting".

5.) Another pop-up appears:

`The system needs to be rebooted for the changes to take effect. Do you wish to reboot now?`

Remove the CD-ROM from the drive.

6.) Click YES. Then, when the system comes up again, wait for about 20-30 seconds, and a pop-up window appears. Click OK.

3.3 Set Time and Date

If the time and date is correct, skip this procedure. Otherwise, you must set the date and time on the OC

1.) Open a Shell on the OC from the toolchest, become root by typing**: su root** **ENTER**.

2.) Type the root password and press **ENTER**. (The password is **#bigguy**)

3.) Type: **setdate *mmddhhmmyyyy*** (e.g.: **setdate 050409001999**) and press **ENTER**.

Note: Alternate method exists	**You may also type: setdate ENTER and you will be prompted through the individual entries.** *Where:* ***mm*** is month (01-12) ***dd*** is day (01-31) ***hh*** is hour (00-23) ***mm*** is minutes (00-59) ***yyyy*** is year (1980-2030)

4.) The **setdate** program will set and display the time and date for on the OC. Verify the date and time on the OC is correct.

5.) **exit** **ENTER** (to exit the root session).

6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.4 Start CT Application Software

To start up the system, type: **startup** **ENTER**

If HIPAA is turned on, you will see a blue login screen covering the CT applications and a pink popup stating "`OC initializing`." After the pink popup allows you to move the cursor out of the box, you will need to login with a user name and password.

Enter the user name as: **service** and the password as: **4rhelp**. Then select LOGIN.

You will have the ability to add your own ***username*** later in the configuration.

Once system state has been saved and restored from previous loads with versions that support this feature, the previous users and passwords will be available.

3.5 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State DVD, skip this section (however, understand that you will have to perform ALL Characterizations and Calibrations in order to make the system operational).

Comment: Image Works should not be displaying any images when you restore System State (to ensure preferences are restored correctly). If you were in Image Works earlier, you should go back and exit to the Browser.

1.) Insert the Save System State DVD into the DVD drive (in the SCSI Tower).

2.) Click the SERVICE DESKTOP button.

Comment: If the Service Key is installed, the user has to press the (1), (2), (3) keys in order to proceed.

3.) Press the PM option from the list.

4.) Press SYSTEM STATE.

5.) Click on ALL. Then click on RESTORE.

6.) When the DVD drive light is OFF, the DVD is ready. Click YES on the pop-up.

Note: If you have problems, see Appendix A - "Error Messages And Troubleshooting", beginning on page 61 for possible error messages and their recovery.

7.) Review the screen output for errors or missing files.

The scroll bar on the right works only when the tool is not busy performing some task. Please be patient, it takes a little while.

Note: **If you should get SAVE/RESTORE SYSTEM STATE ERRORS:**

The Save/Restore System State process reports status on each file specification listed in the `/usr/g/config/SysState.cfg` file. The status may be:

- `Save (or Restore) of {filename} succeeded`
- `Save (or Restore) of {filename} skipped. File not found`
- `FAILED`

A status of `FAILED`, indicates potential problems, `skipped` means the file not found was not required. Please review the following for each file for which you see an error reported.

Some files simply may not exist on the system. If certain tools were never run then certain files may never get created. You may view `/usr/g/config/SysState.cfg` and look for a required field. `req=Y` means the file in question is required. You may also wish to check the system and DVD to verify whether it was actually missing or if there was a permission or owner problem with the file.

Review failures.

8.) **M3 Software Reload**: To restore CSA protocols, follow steps a-d below for GOC2/Linux Console M3 software to GOC2/Linux Console M3 software reload (perform each time):

a.) Do not remove the system state media (DVD) from the drive after the Restore operation.

b.) Open a shell and type the following:

1.) **`mountDVD`** ENTER

2.) `\cp` SPACE `-f` SPACE `/DVD/service_mod_data/system_state/vxtlBackup.tar` SPACE `/data/vxtlBackup.tar` ENTER

3.) `tar` SPACE `Pxvf` SPACE `/data/vxtlBackup.tar` ENTER

4.) `unmountDVD` ENTER

c.) Close the shell.

d.) Verify that the CSA Protocols (Reformat, Volume Viewer, etc.) are restored correctly.

9.) Click on DISMISS.

3.6 Download Flash Firmware

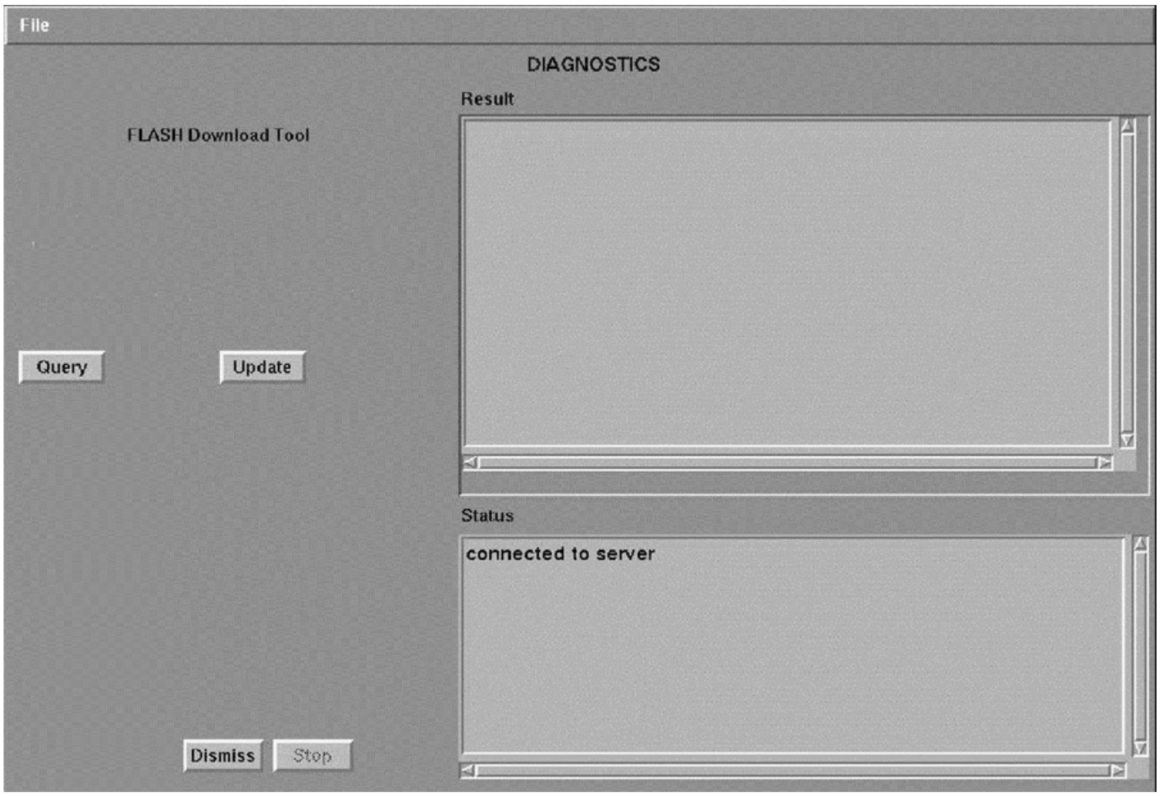

Figure 1-8 Flash Download Tool Graphical User Interface (GUI)

1.) If not already in the Service Desktop, click on SERVICE DESKTOP.

2.) Proceed to Flash Download Tool GUI:

a.) Select UTILITIES menu.

b.) Select INSTALL from the menu.

c.) Select FLASH DOWNLOAD TOOL from the list.

3.) Click UPDATE button to start download.

Note: Ignore all error messages that initially occur. Wait 30 seconds for screen to update.

This will take a few minutes. Any invalid files should be re-downloaded by the software, and the output can be seen on screen.

A message will quickly pop-up, stating: `Download complete without errors`.

4.) Click DISMISS to exit the graphical user interface.

3.7 Adding Software Options

NOTICE Critical option load sequence. **If one of the options to be loaded is "VolumeViewer," then it must be loaded *before* "CT Colonography," "Advanced Vessel Analysis" and "CardIQ2."**

NOTICE for Perfusion and Denta Scan **For systems where CT Perfusion 2 or Denta Scan is included in Available Options, perform this Adding Software Options procedure in the English language environment. (The LFC can be performed in other language environment than English.) Otherwise, the LFC procedure is possible to fail, or if you change the language setting after installing CT Perfusion 2 or Deta Scan, this software option will not work; you will not even be able to start the program.**

Software Options must be loaded every time the OC software is loaded. INSTALL OPTIONS must be executed once, for each option DVD. For example, if you have two options DVDs, you must run INSTALL OPTIONS twice.

Remember, DVDs cannot be interchanged between systems. The first time the DVD is used to install a software option, it must not be write protected. Prior to loading software, the DVD is checked for a valid system ID. **If none exists, the system writes a serial number to the DVD**. The DVD becomes serialized to specific CT systems.

1.) On the Service Desktop, select UTILITIES -> INSTALL -> INSTALL OPTIONS. The `Software Options` Window appears (see Figure 1-9).
2.) Click on the INSTALL button. (see Figure 1-9).

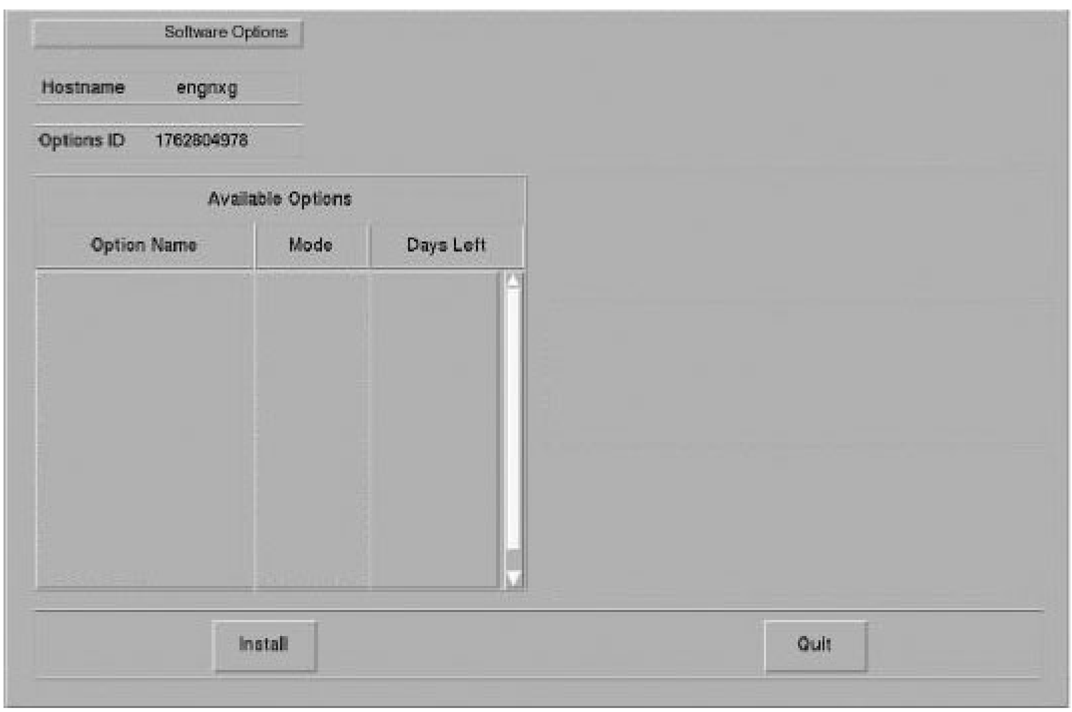

Figure 1-9 Software Options Screen

3.) Click on the PERMANENT button. (See Figure 1-10).

Figure 1-10 Select Mechanism Screen

4.) Click on the MEDIA button (Figure 1-11) and insert the Options Key medium into the SCSI Tower's DVD drive.

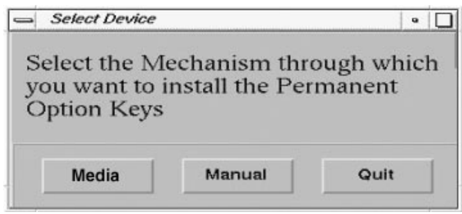

Figure 1-11 Select Device Screen

5.) Click on OK. Software options available for installation are displayed on the options screen.

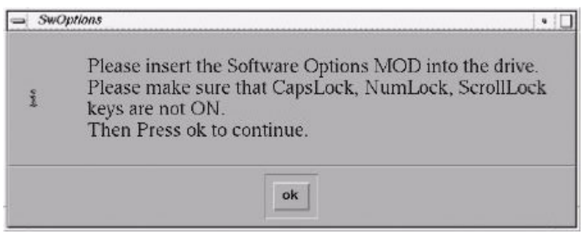

Figure 1-12 SW Options DVD screen

Example: Options screen with Permanent & FlexTrial options.

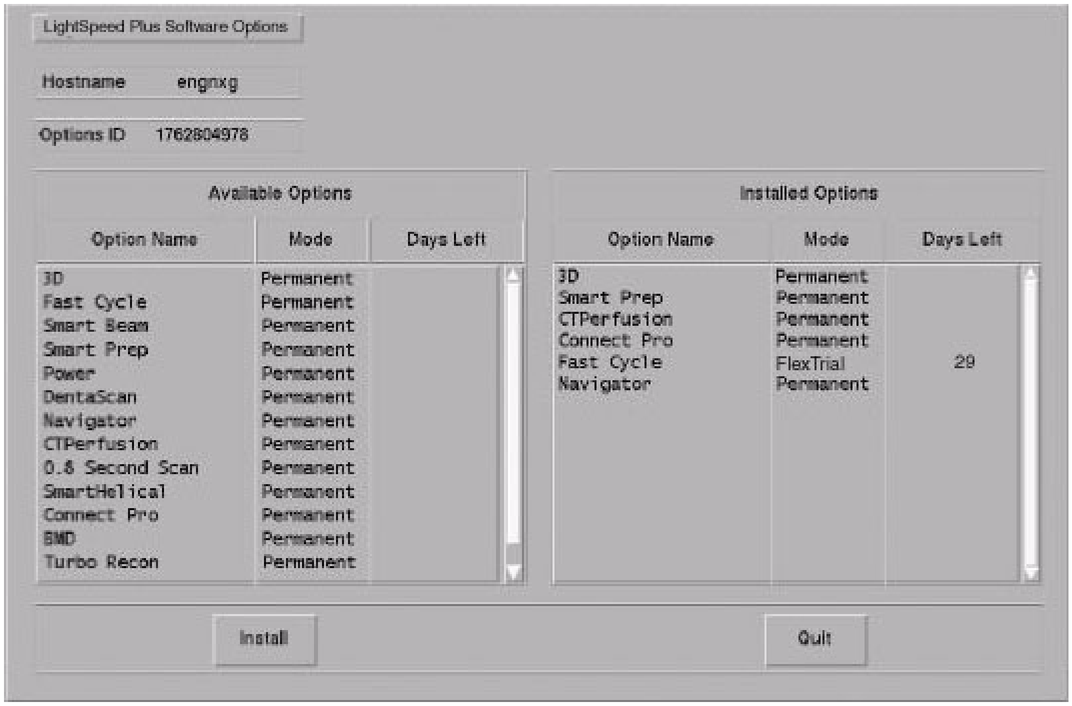

Figure 1-13 Example Options Screen

6.) Select the option(s) to be installed, using your mouse. You can click, drag downwards to select multiple options. Use the scroll bar to reveal options not visible in the window.

7.) Click on the INSTALL button. When the option appears in the "Installed Options" list, the installation of that option is complete. Please be patient, installation time varies by option.

8.) After installing the options, select QUIT twice, then OK. Do not reboot.

9.) Remove the DVD & write protect the side of the DVD with the option installed. The initial install requires the DVD to NOT be write protected; subsequent installs can be done with the DVD write protected). Repeat step this section (Section 3.7) if you have additional options to load.

3.8 Install Cameras

Camera installation is required in the following cases:

- Loading a new system
- When a system state DVD is unavailable, unreadable, or does not contain camera preference info
- If changes need to be made to existing camera configurations

To install cameras, proceed to Chapter 3 - "Laser & DICOM Camera Setup".

3.9 Configure User's System Access

ATTENTION: **The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.**

To prevent potential protocol management issues, a service work-around is required. The work-around, on page 38, must be performed after loading the software patch.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

1.) Press the pink SHUTDOWN button.

2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

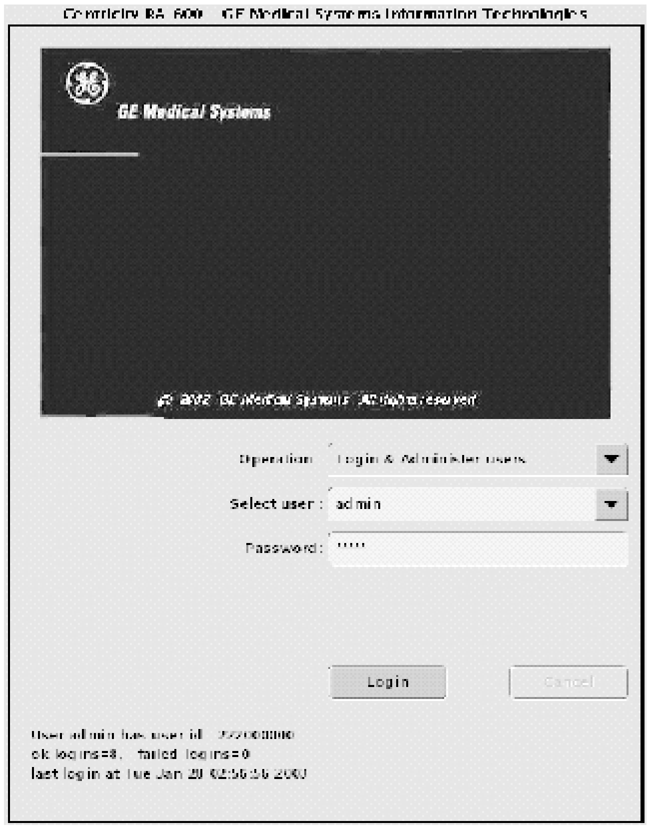

Figure 1-14 Login/Admin Screen.

3.) Select from the pull-down menu `Operation`: Login and Administer Users.

4.) Enter for the `Username`: **`service`** and `Password`: **`4rhelp`**, then select LOGIN.

5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

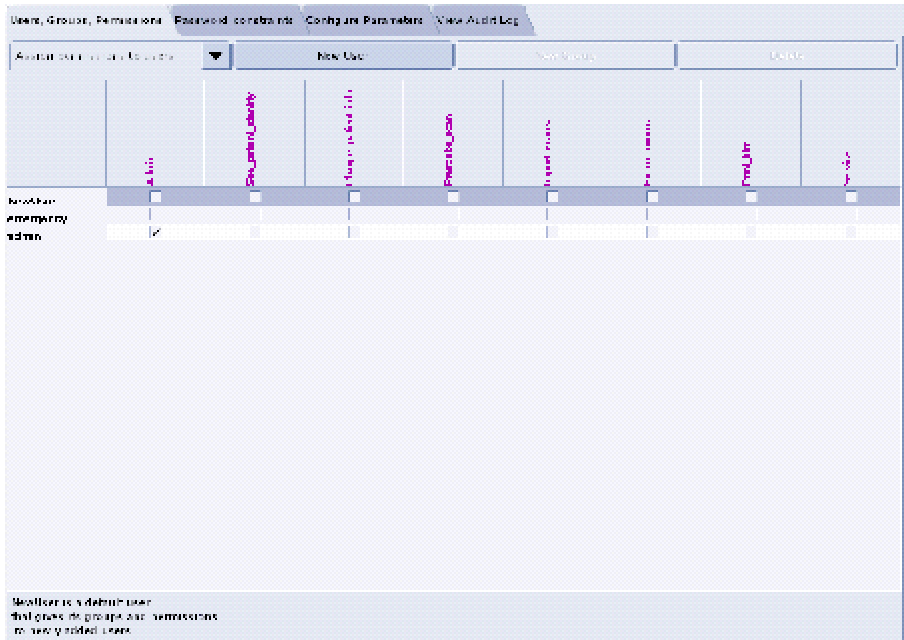

Figure 1-15 User, Groups, Permissions screen

6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:
- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.

7.) Repeat step 6 for as many users as needed. Remember, the admin user may also administer accounts.

8.) If an account for a service user has been added, select `groups to users` from the pull-down menu, and assign the service user to group Service by selecting the check-box under `Service` in the row of the new user name, e.g., John_Q_FE (see Figure 1-16).

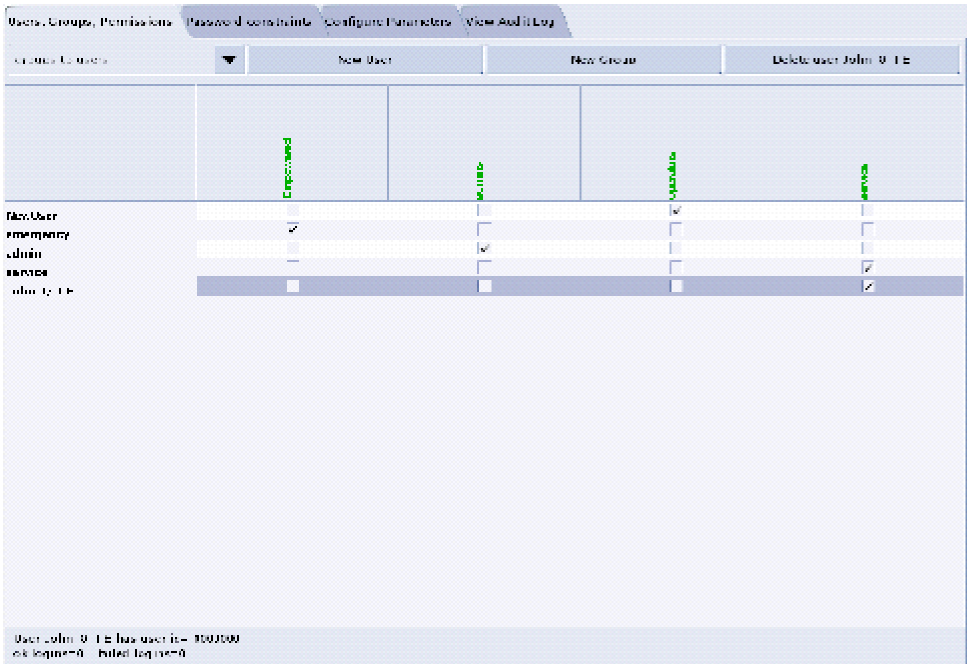

Figure 1-16 Using the User, Groups, Permissions screen

9.) Other users added may also be assigned to one or more groups, depending on their role.

10.) To exit the user administration screens, select the CONFIGURE PARAMETERS tab, and select EXIT ADMIN.

11.) Select from the User name pull-down, the service user name you entered or select the service user. If this is a newly entered user, enter the user name as the initial password, then change and confirm the new password.

3.10 Shutdown System

1.) Press the pink SHUTDOWN button to bring down the system.

2.) Select RESTART and then OK, to bring the system down and back up again.

3.11 Verify System Operation

The following procedure makes sure the system is operating properly after upgrading or reloading the software.

1.) Open a shell and type: **`hinv`**

The output for the OC memory should be at least 2GB. If it is not, check the Host computer to make sure the memory is installed. If it is not installed, order the additional memory.

Note: Check memory	The system must have at least 2GB Memory for proper operation. System performance will be seriously impaired if less than 2GB is present.

2.) ExamRx Desktop

a.) Start the system and make sure you are on the ExamRx desktop.

b.) Verify that the date and time shown in the system status area is correct.

c.) Perform Tube Warm-up and Fastcal.

d.) Set ExamRx Display up to Autoview using your favorite Autoview layout e.) Scan your favorite image quality phantoms to ensure the system is making good images. Make more than one image in a series so you can play with the images below.

f.) Verify that the date and time shown on the images and in the browser are correct.

g.) Watch to make sure the images autoview.

h.) Use List Select to bring an image up in an ExamRx free viewport.

i.) Try each of the Window/Level Preset keys to make sure they are set correctly.

j.) Set the window level via the mouse.

k.) Set the window level via the trackball.

l.) Use the Trackball to Page through the images.

m.) Film at least 1 sheet of film and verify the image quality.

n.) If the site has a Network Image Printer, verify the print quality.

3.) ImageWorks Desktop

a.) Go to the Image Works Desktop.

b.) Select Network and make sure all of the Remote hosts were restored when you restored the system state. If they were not restored, use the information you saved earlier (section 2.2.2 on page 17) to configure the site's networking.

c.) If you have any Genesis based systems configured for networking, and they are up and running, select one of the remote systems and **Query** to see if you can see those systems. *(If the Hospital and the user on a remote system grants you permission, try pushing images to the remote system.)* Use both genesis and non-genesis images if available.

d.) If you can see those systems, try pulling one or more exams onto this system.

e.) If it is all right with your site, try pushing images to all Genesis and DICOM systems that you have in your network configuration.

f.) Use Archive to restore one or more of the most recent Exams performed by the site.

g.) Select an interesting Exam in the Browser and click on CT Viewer to see the images.

h.) Try each of the Window/Level Presets to make sure they are set correctly.

i.) Set the window level via the mouse.

j.) Try a reformat of an interesting series.

k.) If you have the 3D option, try doing a 3D rendering of an interesting series.

l.) Print at least one film and check the image quality.

m.) Remove one or more of the Exams you pulled onto the system.

4.) Service Desktop

a.) Go to the Service Desktop.

b.) Click on TOOLBOX/UTILITIES to see the list of Utility Applications.

c.) Click on TOOLS and then VERIFY SECURITY to see if the system can see your security key.

d.) Click on INSTALL and then VERIFY OPTIONS to see if the right options are configured.

e.) Click on SCAN ANALYSIS to see if you can see the scan files you made earlier.

f.) Click on SYSTEM RESETS, perform a SCAN HARDWARE RESET, and verify that it completes normally.

g.) When the Scan Hardware Reset is complete, click on CLEANUP to exit all the Service applications.

SOFTWARE LOAD COMPLETE. FOR M3 SYSTEMS ONLY, LOAD PATCH (NEXT SECTION).

Section 4.0
Liberty M3 Patch

This section applies only to Linux/Liberty M3 systems. It does NOT apply to M4 systems.

4.1 Load Instructions

1.) Shut down Applications from the service desktop.
2.) Open a shell from the tool chest.
3.) login as root:
    su - root
    password: **#bigguy**
4.) Insert the CD in the drive and mount it:
    mount /mnt/cdrom
5.) Install the patch:
    /mnt/cdrom/install_patch
6.) Un-mount the CDROM:
    umount /mnt/cdrom
7.) Insert the CD in the drive and mount it:
    mount /mnt/cdrom
8.) Install the patch:
    /mnt/cdrom/install_patch_e
9.) Un-mount the CDROM:
    umount /mnt/cdrom
10.) Reboot the system:
    reboot

4.2 Verification

1.) Run the following command:
    showprods | grep PATCH
2.) Output should look something like this:

```
I Helios_Product        /28/2003    LightSpeed ver LINUXM3 rel LINUXM3_PATCH
```

4.3 Service Work-Around

⚠ NOTICE

- **If the Liberty M3 Patch is applied *after* HIPAA is turned ON, then this procedure is not required.**
- **To turn HIPAA ON after the patch is applied, however, the HIPAA setting must be set to OFF (using the steps below) *and then* ON.**

To prevent protocol management issues, perform the following work-around procedure.

1.) Shutdown applications
2.) Open a terminal window

3.) Run Reconfig

Make certain that `HIPAA Present` is set to OFF, and select ACCEPT (even if OFF was previously set).

4.) Reboot the system

Chapter 2
Re-configuration of CT Application SW

Section 1.0
Overview

This chapter describes how to modify an existing system configuration. Many of the parameters entered during a LFC can be changed without a complete re-installation of software. However, to change computer type or image generator hardware requires a re-load of CT applications.

The procedure consists of running the software utility **`reconfig`**, changing system values and accepting the changes. The following table indicates the parameter settings that can be changed using this **`reconfig`**.

SYSTEM SETTINGS	PREFERENCES	HARDWARE SETTINGS	NETWORK SETTINGS
Hospital Name	Screen Display Language	Network Image Printer	Suite Name
Service ID	Preferred Fastcal kV		Host Name/AE Title
Time Zone	Date Display Format		Host IP Address
Next Exam #	Time Display Format		Net Mask
Scan Database Regeneration	Dose Information Display		Broadcast Address
			Default Gateway Address
			NIS (Yellow Pages)

Table 2-1 System Re-configuration Settings

Section 2.0
Procedure

2.1 Overview

In the procedure that follows, you will initialize the reconfiguration utility, modify as needed system setting and commit changes. You can quit the reconfiguration procedure at any time by pressing the QUIT button. Quitting exits the utility without changing the any system setting. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

While the reconfiguration is going on, messages are displayed in a shell window, which closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the file **`/var/adm/install.log.`*****`YYYMMDDWWWHHMMSS`***

Where *`YYYYMMDDWWWHHMMSS`* is the Date/Time that the reconfiguration was started.To view the file, type:**`more /var/adm/install.log.`*****`YYYMMDDWWWHHMMSS`***

2.2 Starting “Reconfig”

1.) If CT Applications is up and running, shutdown system applications to the `ctuser` level.
   a.) Click on the SERVICE DESKTOP button.
   b.) On the desktop toolbar select UTILITIES.
   c.) Select APPLICATIONS SHUTDOWN (to bring down applications only).
2.) On the OC, open a Shell window.
   a.) Type: **`su -`** **ENTER** at the prompt.
   b.) Enter the root (super user) password: **`#bigguy`**
3.) Launch the Install utility:
   Type: **`reconfig`** **ENTER** at the prompt.
   The OC displays the Install Utility Window as shown in Figure 2-1.

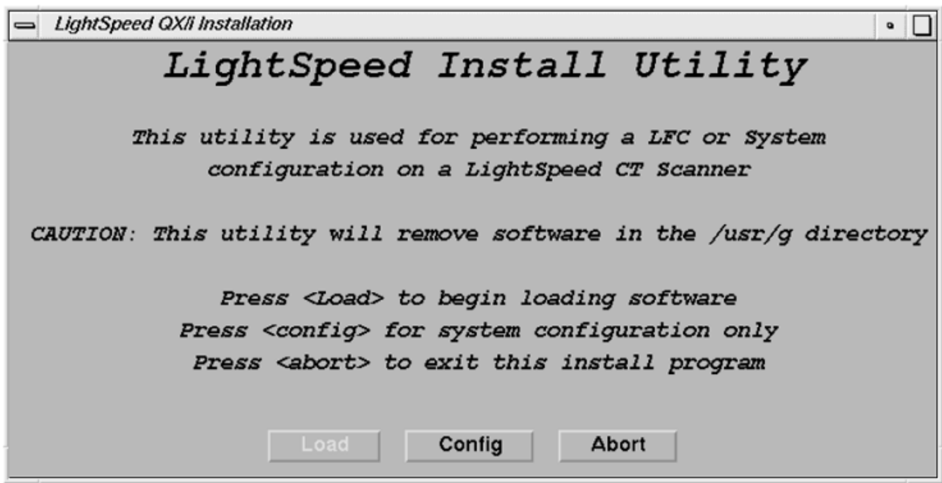

Figure 2-1 Install Utility Window

4.) Using the mouse, click on the CONFIG button.
   The System Configuration “System” Settings Screen is displayed.

2.3 System Configuration Settings

1.) When the System Settings Screen appears (Figure 1-4, on page 23), select either YES or NO to `Regenerate Database`. Select NO to `Regenerate Database`, unless it’s absolutely necessary to rebuild the scan database. This selection determines whether the Scan Database will be regenerated. To avoid accidental “regeneration”, NO is the default selection.

NOTICE Potential for Data Loss	**Select YES if you want to regenerate the database. If YES is selected, then all scan data is deleted (erased) on the Scan Data Disk. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.** **This procedure is normally only needed when a Scan Data Disk is replaced or reformatted.**

2.) For setting up the remaining parameters, follow the instructions in Chapter 1, beginning at Section 3.2.2.2, and then return to step 3 when complete.
3.) After accepting the changes, application software, OS patches, kernel changes take place. The configuration process takes approximately 5 minutes. Status and results are displayed in

a shell window, which closes when the loading process is complete.

Output is logged to a file named: `/var/adm/install.log.YYYYMMDDWWWHHMMSS`

`(Where` *`YYYYMMDDWWWHHMMSS`* is the Date/Time that the loading process was started).

2.4 Finishing System Reconfiguration

1.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See Figure 2-2.)

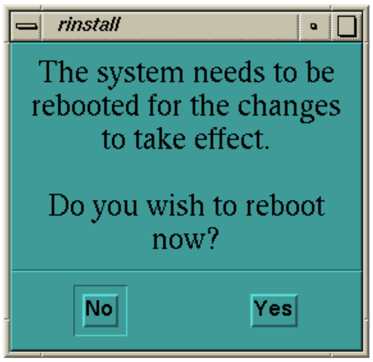

Figure 2-2 Reboot Screen

2.) The system will automatically login as `ctuser` after the reboot. Select OK on the Autostart Disabled popup message.

APPLICATIONS START-UP

3.) In the Console shell, type: **`st`** ENTER

Chapter 3
Laser & DICOM Camera Setup

Purpose: This Chapter illustrates the steps involved to setup a camera. Additionally, data sheets are provided to record the information needed to complete setup.

Section 1.0
Camera Setup

⚠ **NOTICE Potential For Data Loss** **Empty all filming queues before modifying any camera parameter.**

Note: SCSI-1 will not work with DASMs. Use SCSI-2, instead.

1.) Click on the SERVICE DESKTOP button.
2.) On the Desktop Toolbar select UTILITIES **->** INSTALL **->** INSTALL CAMERA**.** The Install Camera window appears, along with a warning message pop-up box that reminds you that all filming queues must be empty before you begin to update or delete a camera.
3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 3-2). First, you must click OK in the warning message box. See Figure 3-1.

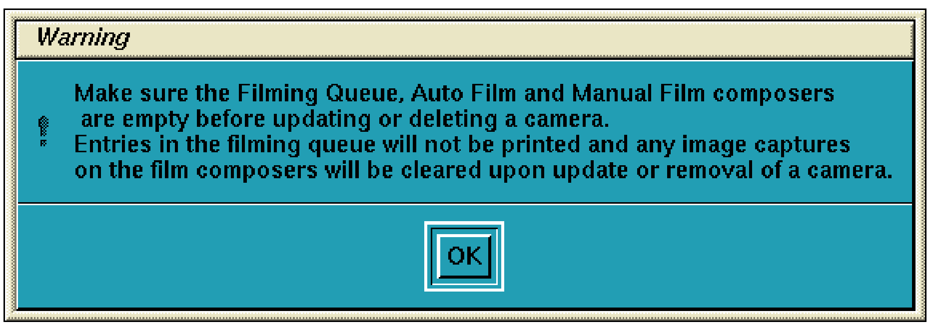

Figure 3-1 Warning Screen

4.) Now you are asked a series of questions.

a.) To add a new camera, click the ADD button (Figure 3-2).

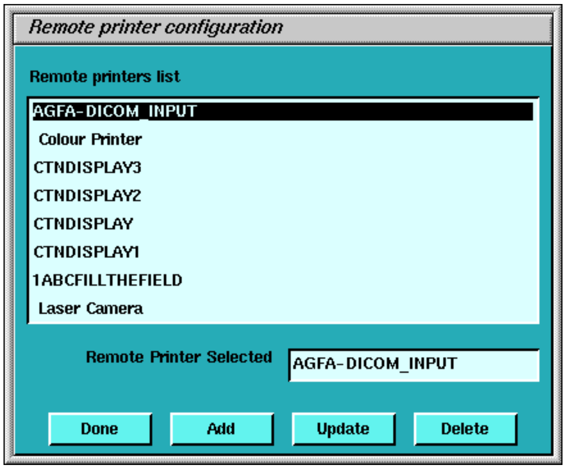

Figure 3-2 Printer Configuration GUI

b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 3-3). If a DASM is present and has not been detected, re-boot the OC and run the camera configuration tool again.

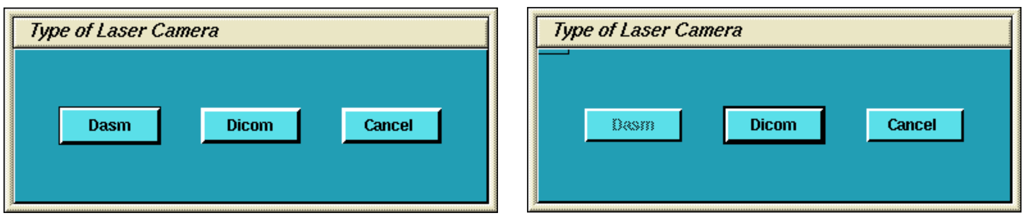

Figure 3-3 Dialog Box for camera Type (with and without DASM)

5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model form the list and click SELECT (Figure 3-4). Now configure it.

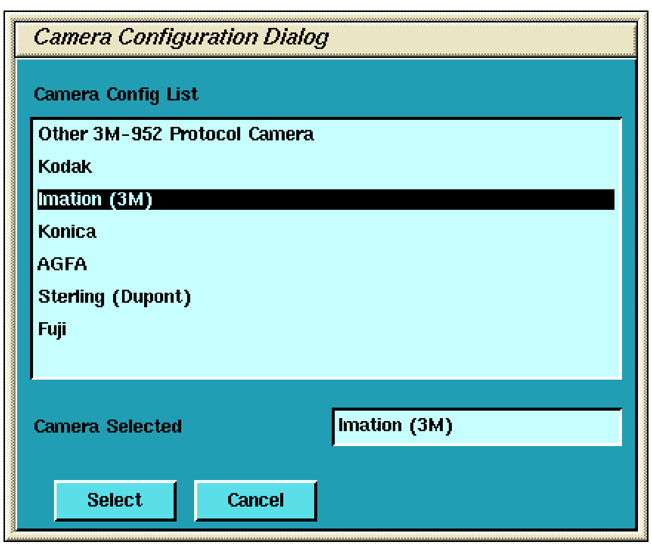

Figure 3-4 Camera Model Dialog (with DASM)

a.) DASM Interface is automatically detected as either Analogue or Digital

b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.

c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak:SMOOTH

Dupont/Sterling:SMOOTH

3M/Imation (Laser Camera):SHARP

3M/Imation (Dry View):SMOOTH

Agfa: SMOOTH

---

d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

When finished setting parameters, click on DONE and proceed to step 8.

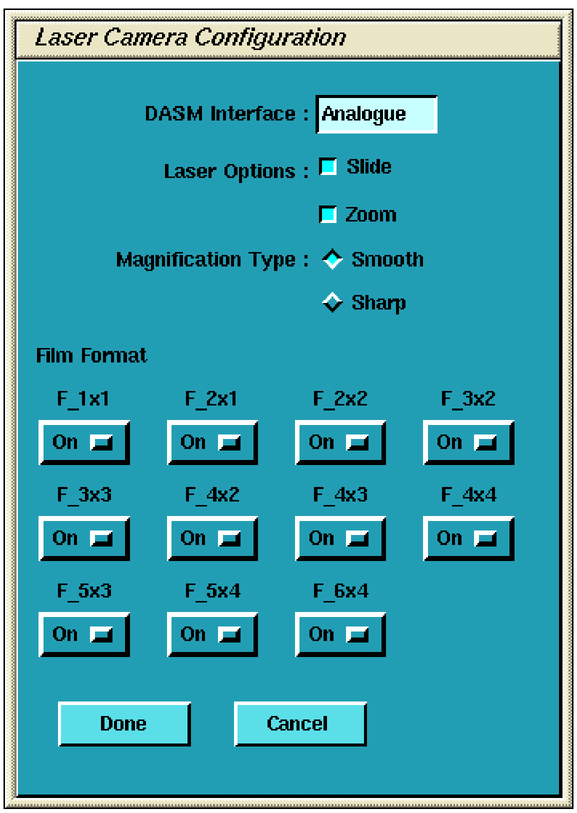

Figure 3-5 Laser Camera Configuration

6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.

a.) A list of camera models appears (Figure 3-6). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

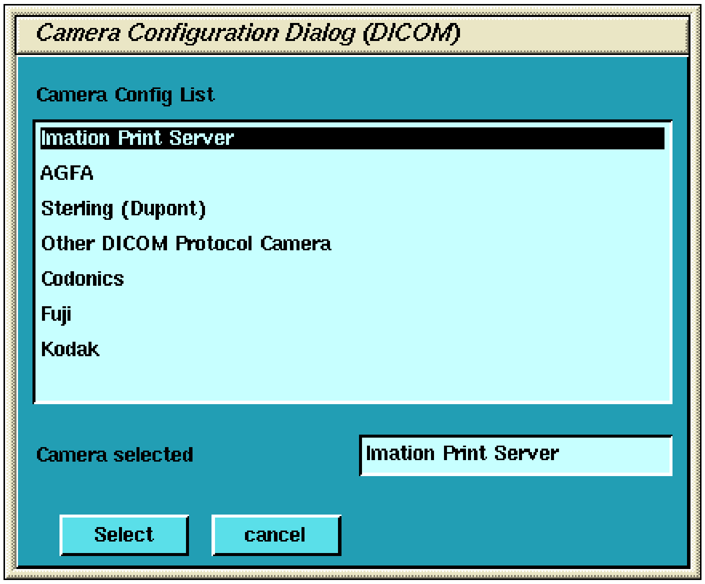

Figure 3-6 Camera Model Dialog (DICOM)

Comment: Re-check the preset info with the camera manufacturer's rep.

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacturer dependent.

b.) Enter the `Network Parameters` (See Figure 3-7)

- > `Device Name` A unique name used to identify the camera.
- > `Host Name` DICOM print server host name, as defined by the hospital.
- > `IP Address` DICOM print server IP address, as defined by the hospital.
- > `AE Title` DICOM print server application entity title, as defined by the print server. *Consult the manufacturer's DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure to check with the camera manufacturer's representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > `TCP/IP Listen Port` DICOM print server TCP/IP listen port, as defined by the server. *Consult the manufacturer's DICOM Conformance Statement.*
- > `Comments` Optional comments used by the DICOM print server.

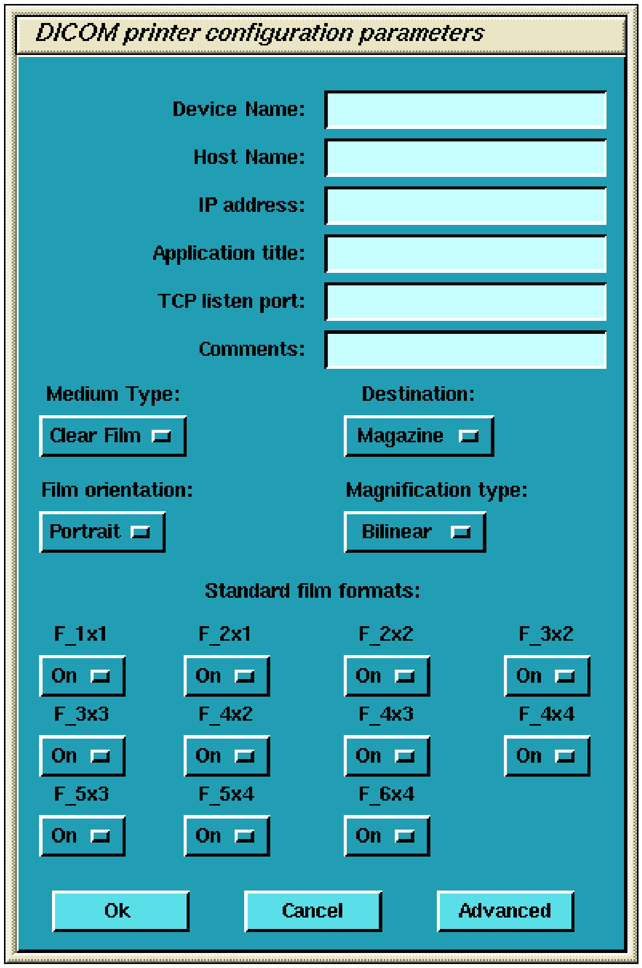

Figure 3-7 DICOM Camera Configuration

c.) `Medium Type specifies` the type of film being used. Currently, only BLUE FILM and CLEAR FILM are suported.

d.) Set `Destination` to the final location for film output. This is either MAGAZINE or PROCESSOR.

e.) `Orientation` selects film orientation. Ony PORTRAIT is currently supported.

f.) Set the `Magnification Type`. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacturer to ensure optimal image quality. Choices are described below:

- > `None` No interpolation. This option is not suported by all cameras.
- > `Replicate` Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal prefrerred.*
- > `Bilinear` A 1st order interpolation of pixels. Results in images usaually described as blurred. *This algorithm is not normal prefrerred.*
- > `Cubic` A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing coefficients" is camera dependent.

Comment: For most camera manufacturers, the preferred selection is CUBIC.

g.) Select the `Standard Film Formats`. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 3-7. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and

AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

h.) After the camera has been configured, click ADVANCED. This automatically creates the camera device file, and pops up the Advanced Parameters screen. See Figure 3-8.

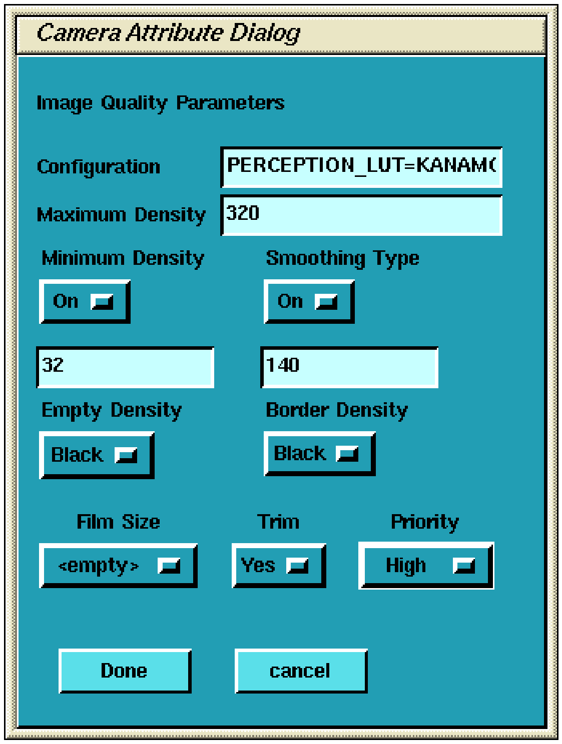

Figure 3-8 Advanced Parameters (Camera Attribute Dialog)

7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative.

A detailed DICOM Conformance Statement for your CT System is available. Refer to this document while working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

a.) `Configuration` - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: Typical Configuration Setting Values:

Agfa Drystar (MG3000)	**PERCEPTION_LUT=KANAMORI (100)**
Imation Dryview (8700)	**LUT=0,7**
Kodak Laser Printer 190	**CS434\CN0\PD1.20**

b.) `Smoothing Type` - Set `Smoothing Type` to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution alogrithm.This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: Recommended Smoothing Type Starting Values and Ranges:
Agfa DryStar (MG3000)Start Value:**140**Range:**137-150**
Imation Dryview (8700)Start Value:**3**Range:**3-13**
Kodak Laser Printer 190Start Value:**Enhanced**Range:**Normal**

c.) `Minimum` and `Maximum Density` - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For `Maximum Density`, input the correct value into the text box. For `Minimum Density`, set it to ON and input the correct value in the text box.

Note: Recommended Minimum and Maximum Density Starting Values:
Agfa Drystar (MG3000)Min.:**20 or 23**Max:**300**
Imation Dryview (8700)Min.:**(Blank)**Max:**300**
Kodak Laser Printer 190Min.:**20**Max:**300**

d.) `Empty Image Density` - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an avialble option. The minimum and maximum density values are used as the representation.

e.) `Border Density` - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.

f.) `Film Size` - Allows the system to specify a particular film size, if selected.

g.) `Trim` - YES produces a white (clear) box surrounding each image.

h.) `Priority` - This sets the print priority.

i.) If you have completed entry of advanced parameters, click DONE.

8.) Go to section 3.10 on page 36 for continuing installation if you are performing a LFC.

9.) **Camera Data Tables:** To locate Install Camera Information: Click on the SERVICE DESKTOP button. On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears. Select each of the cameras that are installed from the list of installed cameras, and click on UPDATE to view the camera's settings. Record the values used to setup each camera in the tables that follow. Extra tables are provided for multiple cameras

Note: You can determine this information by looking at the contents of the following files:
For a DASM Camera: **/usr/g/ctuser/app-defaults/devices/camera.dev**
For a DICOM Print Camera: **/usr/g/ctuser/app-defaults/devices/*name.cfg***
where, ***name.cfg*** is the camera device name from the printer configuration GUI.
Example: **more <*filename from above*>** **ENTER**

End of Procedure

Section 2.0
Data Sheets

Use the following tables to record information about your cameras. Keep for future reference.

DASM CAMERA #1

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	❏ Digital ❏ Analog
Options	Slides or Zoom	❏ Slides ❏ Analog
Film	Smooth or Sharp	❏ Smooth ❏ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 3-1 DASM Camera #1

DASM CAMERA #2

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	❏ Digital ❏ Analog
Options	Slides or Zoom	❏ Slides ❏ Analog
Film	Smooth or Sharp	❏ Smooth ❏ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 3-2 DASM Camera #1

DASM CAMERA #3

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	❏ Digital ❏ Analog
Options	Slides or Zoom	❏ Slides ❏ Analog
Film	Smooth or Sharp	❏ Smooth ❏ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 3-3 DASM Camera #1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES	
DICOM Camera Type	Model Type of Camera		
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.		
Network Parameters	Host Name		
	IP Address		
	AE Title		
	TCP Listen Port		
	Comments		
	Destination	❑ Magazine	❑ Processor
Special Set Up	Orientation	❑ Portrait	❑ Landscape
	Medium Type	❑ Blue	❑ Clear
	Magnification Type	❑ None ❑ Bilinear	❑ Replicate ❑ Cubic
*Advanced Parameters - IQ	Smoothing Type	❑ ON ❑ OFF Value:	
	Configuration		
	Minimum Density	❑ ON ❑ OFF Value:	
	Maximum Density		
	Empty Density (Black/White)	❑ Black	❑ White
	Border Density (Black/White)	❑ Black	❑ White
	TRIM	❑ YES	❑ NO
	Priority	❑ HI ❑ MED	❑ LOW
	Film Size		

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 3-4 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	❏ Magazine ❏ Processor
Special Set Up	Orientation	❏ Portrait ❏ Landscape
	Medium Type	❏ Blue ❏ Clear
	Magnification Type	❏ None ❏ Replicate ❏ Bilinear ❏ Cubic
*Advanced Parameters - IQ	Smoothing Type	❏ ON ❏ OFF Value:
	Configuration	
	Minimum Density	❏ ON ❏ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	❏ Black ❏ White
	Border Density (Black/White)	❏ Black ❏ White
	TRIM	❏ YES ❏ NO
	Priority	❏ HI ❏ MED ❏ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 3-5 DICOM Camera #1

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES	
DICOM Camera Type	Model Type of Camera		
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.		
Network Parameters	Host Name		
	IP Address		
	AE Title		
	TCP Listen Port		
	Comments		
	Destination	❏ Magazine	❏ Processor
Special Set Up	Orientation	❏ Portrait	❏ Landscape
	Medium Type	❏ Blue	❏ Clear
	Magnification Type	❏ None ❏ Bilinear	❏ Replicate ❏ Cubic
*Advanced Parameters - IQ	Smoothing Type	❏ ON ❏ OFF Value:	
	Configuration		
	Minimum Density	❏ ON ❏ OFF Value:	
	Maximum Density		
	Empty Density (Black/White)	❏ Black	❏ White
	Border Density (Black/White)	❏ Black	❏ White
	TRIM	❏ YES	❏ NO
	Priority	❏ HI ❏ MED	❏ LOW
	Film Size		

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 3-6 DICOM Camera #1

Chapter 4
Changing System Time Zone

Reconfig has the ability to change the system's time zone setting without having to perform a complete LFC. The following procedure can be followed when the system is installed in a location where there is no local time zone selection available. (See "Set Time and Date" on page 28.)

1.) If Applications is running, shutdown Applications first by selecting UTILITIES -> SHUTDOWN APPLICATIONS from the SERVICE DESKTOP.
2.) Open the console window, by double-clicking on the console icon in the task box.
3.) In the console window, type **`su -`** and press **ENTER**
4.) Enter the root (super user) password: **`#bigguy`**
5.) Now type: **`reconfig`** **ENTER**
6.) Select CONFIG from the installation title screen
7.) On the System Configuration screen, select the Time zone for this site. It's suggested that you record it below for future use:

Region / Time Zone

Table 4-1 Time Zone Setting

8.) Press ACCEPT to accept re-configuration
9.) Now click YES to reboot the system.
10.) Proceed to Set Time and Date on page 28 to setup time on OC.

Chapter 5
Regenerating a Scan Database

NOTICE Potential for Data Loss

This procedure removes all scan and calibration data from the system disks. Make sure that all Images have been reconstructed before performing this procedure.

This procedure will re-initialize the Scan Data Disk, erasing all scan and calibration data. You will need to restore Calibrations (from System State), or recalibrate the system once you have performed this procedure.

Reconfig has the ability to regenerate the Scan Database. The initial OC reconfiguration screen (Figure 1-4, on page 23) defaults the `Regenerate Database` selection to NO.

This procedure will most likely be needed/used when the Scan Data Disk has been replaced or reformatted. Using this procedure will eliminate the need to perform a complete LFC in these circumstances.

1.) If you need to regenerate the Scan Database, perform the Re-configuration of CT Application SW on page 41, and select the YES button for `Database Regeneration` on the System Settings Screen.

2.) Select ACCEPT, and allow the Reconfiguration to complete.

3.) Once the Reconfiguration has completed, and the system is rebooted, reload System Calibrations from the System State DVD, or if the State DVD is not available, perform all system Calibrations.

Appendix A
Error Messages And Troubleshooting

A table of potential error messages and potential solutions is provided below.

SECTION	ERROR	POSSIBLE CAUSE	SUGGESTED RECOVERY SOLUTION
Install CT Application Software on page 28	“pflash” failed to download.		A.) The system must be shut down and restarted prior to manual flash or reconfig. B.) Manually flash the RIP Board firmware. At the `ctuser` prompt, type: **`/usr/g/ice/bin/pflash`** ENTER If the RIP Board firmware flash routine operates correctly, the screen displays firmware loading type text in approximately 12 lines. If the routine continues to fail: - Shutdown the system. - Open the Console to gain access to the VME Chassis (right side of console). - Check the seating of the RIP Board and SDC Board in the VME Chassis. - Power up the system. - Repeat the manual RIP Board pflash process. C.) Manually reconfigure the Scan Data Disk. At the `ctuser` prompt, type: **`/usr/g/scripts/reconfigScanDisk`** ENTER

Restore System State on page 29	`Corruption detected on the MOD in the drive.`
	`MOD cannot be mounted.`
	`MOD initialization completed with failure`
	`Save/Restore System State: Completed with failures.`
		Bad or corrupted MOD.	Attempt to repair MOD with **`fsck`** and reissue command. Open a new window, insert MOD in drive and type (as root):
			`fsck -n /dev/dsk/dks1d3s7` ENTER
			If **`fsck`** fails or system hangs, the system state cannot be loaded from the MOD and the system must be re-calibrated and manually reconfigured. Use re-configuration data recorded in Section 2.2.2
			If **`fsck`** completes successfully, restart procedure at Section 3.5

	CD or DVD will not come out of drive	Not unmounted	Unmount the drive, and press the button on the front of the drive. Unmount command is: **`unmount /mntcdrom`** *<-HostPCDVD* **`unmount /mnt/cdrom1`** *<-SCSITowerCD* **`unmount /mnt/cdrom2`** *<-SCSITowerDVD*

Table 5-1 Potential LFC Error Messages and Recovery Solutions

GE MEDICAL SYSTEMS

GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.

GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPOR